



Uncertainty quantification by large language models

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ABSTRACT

As reasoning capabilities of large language models (LLMs) continue to advance, they are being integrated into increasingly complex scientific workflows, with the goal of developing agents capable of generating evidence-based explanations and testing hypotheses and theories. However, despite their rapid progress, most existing evaluations of LLM reasoning focus on accuracy or consistency rather than on uncertainty quantification (UQ), which is essential for evidence-based reasoning because it quantifies the degree of trustworthiness of evidence-based explanations. Current approaches to LLM uncertainty remain fragmented, often lacking standardized benchmarks that test models under varying task complexities. To address this gap, we introduce the first benchmark suite designed to evaluate UQ by LLM-based agents and tools. The benchmark targets one of the most fundamental UQ problem: estimating whether one quantity is probably larger than another under uncertainty. It includes two progressively complex tasks: a simple inequality test, where models judge whether one of two sets of samples is “larger,” “smaller,” or “uncertain” with 95% confidence, and a complex inequality test, where models assess interventional probabilities requiring multiple intermediate calculations. We found that reasoning models are generally capable of UQ (scores $\geq 70\%$) in the simple inequality case but do not score appreciably better than random guessing (scores $\sim 33\%$) for the complex inequality case if the UQ method and intermediate steps are not provided in the prompt. Our implementation is available at <https://github.com/bekaiser-LANL/tether>.

1. Introduction

Recent breakthroughs in large language modeling have revitalized aspirations for artificial intelligence (AI) agents capable of autonomously supporting scientific research (Lu, Aleta, Du, Shi, & Moreno, 2024; Watkins, 2024), including agents capable of literature review, data collection, peer review, and quantitative data analyses (Messeri & Crockett, 2024). LLMs have successfully performed subject-specific tasks for scientific applications in chemistry, biology, medicine, and physics (Agathokleous, Saitanis, Fang, & Yu, 2023; Ahmad, Öfverstedt, et al., 2024; Guo et al., 2023; Hong, 2024; Nguyen et al., 2023). LLMs have demonstrated zero-shot learning capabilities, motivating their continued use as a subject-matter expert’s assistant (Brown et al., 2020; Hong, 2024). Many LLMs now possess the kinds of basic mathematical and arithmetic capabilities required for rudimentary data analyses (Chern et al., 2023; Luo et al., 2025; Xia, Li, Liu, Wu, & Liu, 2025; Yu et al., 2024). Statistical analysis benchmarks have also assessed LLM abilities to perform basic calculations and tests given specific datasets (Zhu, Du, Li, Luo, & Tang, 2024). High-stakes scientific research demands that these capabilities be developed in tandem with methods for assessing model/agent trustworthiness, as

shown by recent multilingual knowledge-graph alignment work emphasizing uncertainty-aware inference (Akhtar et al., 2024). Similar to how automatic segmentation frameworks in large-scale CT imaging require uncertainty quantification to ensure the reliability of voxel-wise body composition analyses, as well as in transfer learning-based histopathological and voxel-wise CT registration studies that address hierarchical and propagation uncertainties (Ahmad, Asghar, & Gillani, 2022; Ahmad, Dahlberg, et al., 2024; Ahmad et al., 2023), LLM-based scientific agents must likewise integrate UQ to provide trustworthy quantitative reasoning.

Unlike formal mathematical proofs and deductive reasoning, UQ is an explicitly inductive task that accounts for the platonic imperfection of the ‘real world’ (Sullivan, 2015). UQ involves problem-specific, subjective judgments grounded in the characteristics of the data (e.g., the treatment of outliers, knowledge or hypotheses of variable dependencies, etc.), discipline-specific standards (e.g., $4-5\sigma$ levels of significance are required for the acceptance of a discovery in exoplanetary science (Madhusudhan et al., 2025), and may depend upon ulterior concerns such as high-stakes policy decisions. Arguably, the benchmarks presented here are perhaps best viewed as quantitative indicators of typical UQ behavior, rather than as evaluations of deductive reasoning.

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Assessing trustworthy UQ behavior is a unique challenge because it is inherently ambiguous—driven by a wide range of subjective methodological choices. This ambiguity complicates the task of determining whether a model’s approach to UQ is reliable or trustworthy. Therefore, it is critical to characterize both the subjective UQ choices that models make autonomously and how well they perform when prompted to use specific UQ techniques, while remaining robust to prompt variability. Moreover, trustworthy UQ is a valuable benchmark for AI development because scientific rigor improves AI performance (Pineau et al., 2021).

We propose a model-agnostic suite of benchmarks for LLMs and LLM-based agents focused exclusively on the most elementary form of UQ: the estimation of empirical inequalities under uncertainty. Before addressing complex, noisy, and ambiguous real-world problems, a model should demonstrate competence on simple UQ tasks. Strong performance at simple UQ tasks is a necessary, though not sufficient, criterion for establishing UQ trustworthiness.

The proposed benchmark suites do not adjudicate whether models understand UQ in a human-like sense, but rather they assess how well model behavior aligns with common subjective UQ methodological choices (e.g., *a priori* use of *t*-distribution) and specific UQ instructions. While the distinction between mere pattern-matching and true understanding is important for qualitative tasks, it is less important in quantitative domains like UQ because the alignment of model outputs with trustworthy results can be directly assessed. Moreover, human understanding itself is rarely, if ever, judged by inspecting internal cognitive processes. Instead, it is typically assessed through behavioral means (Barman, Caron, Claassen, & De Regt, 2024).

The novel contributions of this study are as follows:

- While numerous studies investigate UQ of LLM outputs (Abdar et al., 2021; Huang et al., 2025; Nikitin, Kossen, Gal, & Martinen, 2024; Tanneru, Agarwal, & Lakkaraju, 2024; Vashurin et al., 2024; Ye et al., 2024; Zhang et al., 2023), to our knowledge, the benchmark suite proposed here is the first to measure a model’s ability to *perform* UQ.
- UQ of LLM output is essential for safety and reliability (Hong, 2024; Huang et al., 2024), hence we recommend and demonstrate UQ for model performance on the proposed UQ benchmark.
- We use a Multiple Choice Question (MCQ) format to reduce lexical (word choice), syntactic (word order), and semantic (word meaning) uncertainties (Nikitin et al., 2024) in a model-agnostic manner.
- Because the proposed benchmarks are randomly generated, aleatoric model uncertainty can be reduced by setting the number of benchmark questions N_q to a large number, e.g., $N_q \geq 180$.

In Section 2 the importance of estimation of empirical inequalities under uncertainty through history is outlined. Section 3 discusses the composition of the benchmark suite, including the prompts and data used to test each model, Section 4 presents the benchmark results for all of the models tested, and Section 5 presents the conclusions from this study.

2. UQ of empirical inequalities are fundamental in science

The estimation of empirical inequalities under uncertainty is fundamental to science, and many scientific breakthroughs have hinged on the estimation of empirical inequalities under uncertainty. Einstein’s theory of general relativity superseded Newtonian theory when Eddington measured 1.98 ± 0.16 arcsec deflection of starlight around the sun because the lower bound of Eddington’s confidence interval is closer to the relativistic prediction of 1.75 arcsec (Einstein et al., 1911) compared with Newtonian prediction of 0.875 arcsec (Von Klüber, 1960). The debate surrounding the relative importance of smoking versus genetic predisposition for developing lung cancer raged for a hundred years before Cornfield argued that the observed link between

smoking and lung cancer could not plausibly be explained by an unmeasured confounder unless it had implausibly high associations with both smoking and cancer. His method for estimating how significant an unmeasured confounder must be to dismiss an observed association between an intervention and an outcome is now known as Cornfield’s inequality (Greenhouse, 2009; Pearl & Mackenzie, 2018). Recently, astronomers inferred the presence of dimethyl sulfide (DMS) and/or dimethyl disulfide (DMDS) by comparing the spectral features of the mid-infrared transmission spectrum of the atmosphere of the exoplanet K2-18b with the corresponding features of DMS and DMDS spectra. Since DMS and DMDS are associated with aquatic life on Earth, the result implies the possible discovery of life on another planet. Specifically, they found that the spectral peak magnitudes associated with K2-18b’s atmosphere and the molecules overlapped within roughly the 99.7% (3σ) confidence interval (Madhusudhan et al., 2025). In these examples, the result hinged on quantitative confidence that the magnitude of one inferred or observed variable is larger than a previous theoretical prediction, another inferred variable, and zero, respectively.

3. A suite of UQ benchmarks

The suite consists of 10 benchmarks, each of which generates synthetic empirical inequality data randomly and then prompts the models to estimate the inequalities from the synthetic data with a target confidence level of 95%. A 95% confidence level was arbitrarily selected because it is a commonly adopted *a priori* subjective convention in scientific practice (Swift, 2009). Variations of the benchmark suite that increase or lower the confidence level could also be implemented.

In four of the benchmarks, the model is free to choose the UQ method. In another four of the benchmarks, we specify a UQ method in the prompt. In the remaining two benchmarks we include ontology instead of abstract variables in the prompt. In the parlance of quantum mechanics, model results from prompts that specify the UQ method indicate how well a given model can ‘shut up and calculate’ (Georgescu, 2022). Model results from prompts that include ontology indicate the effects of model awareness of well-known results derived from empirical inequalities under uncertainty (e.g., *a priori* knowledge that smoking causes cancer). In half of the benchmarks, the models are graded relative to answer choices informed by confidence intervals derived from student’s *t*-distribution and in the other half they are graded relative to answer choices informed by confidence intervals derived from bootstrapping, using 1000 bootstrap iterations.

A MCQ format was used for all questions to be consistent with established practices in natural language processing model evaluations (Liu, Cao, Liu, Ding, & Jin, 2024; Ye et al., 2024), for unambiguous grading, and to permit comparison with random guessing. The number of questions N_q is a free parameter, where $N_q \geq 9$ and N_q must be a multiple of 9 because the benchmark generator produces benchmarks with equal proportions of answers $A = \text{yes}$ with 95% confidence, $B = \text{no}$ with 95% confidence, $C = \text{uncertain}$ and evenly subdivides these sets of answers into “easy”, “medium”, and “hard” difficulty. To generate inequalities of differing difficulty, they are generated evenly in three bins of absolute difference magnitudes between the two unequal variables, $|\Delta|$ (Table 1). For Simple Inequality, the two unequal variables are means of two vectors of samples of normally distributed variables. The **Simple Inequality** benchmarks measure a model’s ability to infer whether it is more probable that one variable is greater than another variable when provided 10 samples of each variable. For Complex Inequality, the two unequal variables are two ‘do’ probabilities calculated from a frequency table. The **Complex Inequality** benchmarks increase the complexity of arithmetic calculations and measure the model’s ability to infer causal relationships (or lack thereof) given data from a frequency table of binary cause, effect, and mediator variables.

The bins of $|\Delta|$ loosely correspond to the notion of problem difficulty because (holding sample size constant) when the $|\Delta|$ is large, the correct choice of A , B , or C can still be obtained by using different UQ methods

a. Prompt example for Simple Inequality with unspecified method

Vector 1: {vector 1} Vector 2: {vector 2}

Is it more probable that a sample from Vector 1 is greater than sample from Vector 2? Answer 'A' for yes, 'B' for no, or 'C' for uncertain. Use only the data provided here and the 95 percent confidence level. Do not repeat the prompt. Answer:

b. Prompt example for Simple Inequality specified method (bootstrap)

Vector 1: {vector 1} Vector 2: {vector 2}

Is it more probable that a sample from Vector 1 is greater than sample from Vector 2 Use bootstrap resampling to calculate the 95 percent confidence levels. Use the 95 percent confidence intervals to answer 'A' for yes, 'B' for no, or 'C' for uncertain. Use only the data provided here and the 95 percent confidence level. Do not repeat the prompt. Answer:

c. Prompt example for Complex Inequality with unspecified method and specified ontology

Consider the following causal inference problem. The number of samples that do not smoke, do not have lung cancer, and do not have tar deposits in lungs is {number 1}. {number 2} samples do not smoke, do not have lung cancer, and do have tar deposits in lungs. {number 3} samples do not smoke, do have lung cancer, and do not have tar deposits in lungs. {number 4} samples do not smoke, do have lung cancer, and do have tar deposits in lungs. {number 5} samples do smoke, do not have lung cancer, and do not have tar deposits in lungs. {number 6} samples do smoke, do not have lung cancer, and do have tar deposits in lungs. {number 7} samples do smoke, do have lung cancer, and do not have tar deposits in lungs. {number 8} samples do smoke, do have lung cancer, and do have tar deposits in lungs. Does smoking cause cancer? Use the front-door criterion to determine if smoking causes cancer from the provided data. Use the standard error of proportion and t distribution on the final front door probability difference to calculate the 95% confidence level intervals. Use the 95% confidence levels to answer 'A' for yes, 'B' for no, or 'C' for uncertain. Please use only the data provided here.

Fig. 1. Examples of prompts. The bold items in curly brackets refer to randomly generated vectors of length 10 (Simple Inequality) and the 8 integer frequencies (Complex Inequality, Table 2 samples column $n_1, \dots, n_8$) that are substituted into the prompts.

Table 1

Difficulty bins are used to ensure that the synthetically generated inequalities evenly span the absolute difference magnitudes $|\Delta|$ for all A , B , and C answers. Δ_{XY} is the difference between sample means and Δ_p is the difference between 'do' probabilities.

Difficulty	Simple inequality	Complex inequality
Easy	$1.33 < \Delta_{xy} \leq 2.00$	$0.25 < \Delta_p \leq 1.00$
Medium	$0.66 < \Delta_{xy} \leq 1.33$	$0.05 < \Delta_p \leq 0.25$
Hard	$0.00 < \Delta_{xy} \leq 0.66$	$0.00 < \Delta_p \leq 0.05$

because there is greater numerical tolerance. Hence a large $|\Delta|$ problem is “easy”. As $|\Delta| \ll 1$, the choice of A , B , or C is less likely to align with the benchmark solution unless the same UQ method is used, hence the problem is “difficult”. The difficulty bins also ensure that nearly overlapping unequal variables are not overrepresented in C questions. The even partitioning of the difficulty bins for all A , B , and C solutions ensure reproducibility by ensuring that all questions span the same range of $|\Delta|$, and the even partitioning of the benchmark into A , B , and C solutions (Appendices A.2.3 and A.3.4) ensures that the grade for random guessing converges to 33% in the limit of $N_q \gg 1$. Finally, the generated questions are randomly shuffled.

3.1. Prompting strategy

We define a successful UQ assistant as one that is able to perform trustworthy UQ analyses in a manner that is robust with respect to prompt strategy. As we show later, this definition of success is supported by our benchmark demonstration because higher quality models tended to perform better. We implemented fixed prompt text

for the Simple Inequality and Complex Inequality benchmarks while the numerical information was varied, as shown in the examples Fig. 1. The benchmarks were implemented with three categories of prompt strategy:

(1) *unspecified method versions of simple inequality and complex inequality benchmarks.* The model is given synthetic data and prompted to answer the MCQ without any further details other than the answer should be calculated for 95% confidence (Fig. 1a). The model must make judgments regarding numerical precision, UQ method, etc.

(2) *specified method versions of simple inequality and complex inequality benchmarks.* The model is prompted with the UQ method in the Simple Inequality benchmarks and prompted with the UQ method and front-door criterion (Pearl, 2009) causal inference calculation method in the Complex Inequality benchmarks. Comparison with the unspecified-method benchmarks isolates for the effect of specified UQ method on model performance.

(3) *unspecified-method with specified ontology version of the complex inequality benchmark.* The prompt contains ontology ('smoking' as the cause, 'tar in lungs' as the mediator, 'lung cancer' as the effect), but not the UQ method nor front-door criterion. Comparison with the unspecified-method benchmarks isolates for the effect of ontology and prior knowledge on model performance: The B and C answers that compose 66% of the questions conflict with real-world understanding of the relationship between smoking and lung cancer.

3.2. Simple inequality benchmarks

The Simple Inequality benchmarks ask models: Which directly observed variable is larger with 95% confidence? Two vectors of samples

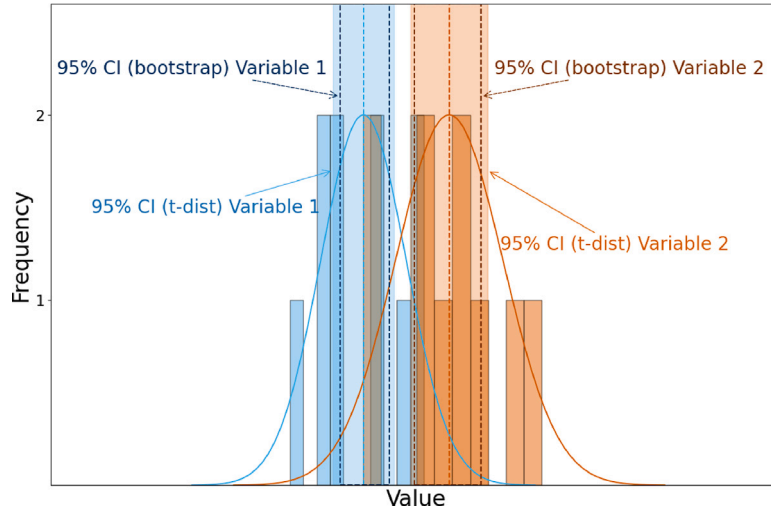


Fig. 2. Example of randomly generated population and sample statistics for a Simple Inequality bootstrapping question. Bootstrapped confidence intervals (dashed lines) and t -distribution confidence intervals (shaded regions) for variable 1 (blue) and variable 2 (orange). The histogram bins contain the samples provided in the vectors in the prompt.

X and Y (both of length N_{xy}) of random variables are sampled from Gaussian distributions (Fig. 2). For each problem, the means of the two Gaussians are randomly selected from a uniform distribution over $[-1, 1]$ and the standard deviations are randomly selected from a uniform distribution over $[0.5, 8]$. Confidence intervals are calculated for the difference between the sample means, $\Delta_{xy} = \bar{X} - \bar{Y}$ (details provided in Appendix A.2) (Cumming, 2009).

3.3. Complex inequality benchmarks

The Complex Inequality benchmarks asks: Which inferred variable is larger with 95% confidence? The inferred variables require a sequence of calculations (greater arithmetic complexity) and are two ‘do’ probabilities, $P(Y = 1 | \text{do}(X = 1))$ and $P(Y = 1 | \text{do}(X = 0))$. The questions are constructed using the front-door criterion (Pearl, 2009) to define the relevant ‘do’ probabilities,

$$P(Y | \text{do}(X)) = \sum_Z \left(P(Z | X) \sum_{X'} P(Z | X', Z) \cdot P(X') \right), \quad (1)$$

where X , Y , and Z are binary variables that correspond to cause, effect, and mediator. Answer choice *A* corresponds to ‘ X causes Y ’ through the mediator Z (i.e., $P(Y = 1 | \text{do}(X = 1)) > P(Y = 1 | \text{do}(X = 0))$) with 95% confidence, answer choice *B* corresponds to ‘not X causes Y ’ (i.e., $P(Y = 1 | \text{do}(X = 1)) < P(Y = 1 | \text{do}(X = 0))$) with 95% confidence, and answer choice *C* corresponds to the cases where neither inequality of the variables is supported by the evidence with 95% confidence.

The confidence intervals for the inequalities are calculated on the difference between $P(Y = 1 | \text{do}(X = 1))$ and $P(Y = 1 | \text{do}(X = 0))$. Frequencies $n = \{n_1, n_2, \dots, n_8\} \in \mathbb{N}$ are randomly generated such that all frequencies are greater than or equal to one $\forall i, n_i \geq 1$. $P(Y = 1 | \text{do}(X = 1))$ and $P(Y = 1 | \text{do}(X = 0))$ are calculated by calculating the marginal, joint, and conditional probabilities on the right hand side of Eq. (1) using the frequency data (details provided in Appendix A.3.1). Confidence intervals are calculated for the difference between the ‘do’ probabilities means, $\Delta_p = P(Y = 1 | \text{do}(X = 1)) - P(Y = 1 | \text{do}(X = 0))$ (details provided in Appendix A.3) (Cumming, 2009).

4. Benchmark demonstration

We selected a diverse sample of models that span reasoning, non-reasoning, and fine-tuned for mathematics models. The models tested in this study are Claude 3.5 Sonnet*, Claude 3.7 Sonnet*, DeepSeek-R1-Distill-Qwen-32B, GPT-4.1*, Granite-3.2 (8B), Llama-3.2 (1B), Mathstral (7B), Mistral (7B), o3*, o4-mini*, Phi4 (14B), Qwen2-math (7B),

Table 2

Frequency table for binary variables X , Y , and Z . The number of samples N_s used for computing the sample standard error of proportion for the probabilities is $N_s = \sum_{i=1}^8 n_i$.

X	Y	Z	Samples
0	0	0	n_1
0	0	1	n_2
0	1	0	n_3
0	1	1	n_4
1	0	0	n_5
1	0	1	n_6
1	1	0	n_7
1	1	1	n_8

and WizardMath (7B). We tested LLMs with parameter counts ranging from 7 billion to an estimated 1.76 trillion, the latter being a speculative figure for GPT-4, to provide an assessment of models that are readily available to users with varying computing resources. Models marked with an * indicate unknown parameter sizes since this data is not publicly available. Models available through Ollama were run locally on desktops. OpenAI and Claude models were run through their APIs and DeepSeek-R1-Distill-Qwen-32B was run on four GPUs. Experimental runtime for the Simple Inequality benchmarks was ~ 2 h on average and Complex Inequality benchmarks ran on average for ~ 3 h. The 10 benchmarks we generated for our demonstration can be found in our online repository.

Model outputs were saved and analyzed using a pattern-recognition deterministic grader alongside an LLM grader (Phi4-14B). The LLM was used to confirm or deny that the pattern-recognition algorithm detected the correct multiple choice answer based on its own answer. LLM responses that caused the LLM and deterministic graders to disagree are flagged for human review.

We analyzed results by overall grade and answer choices to understand the full scope of each model’s ability. Across all of the benchmarks, we observed that o4-mini, o3, Claude-3.7, and GPT4.1 outperformed all of other LLMs and, in all cases, scored better than the grade equivalent to random guessing (33.3%). Table 3 presents the grades assigned to each model given each Simple Inequality benchmark. These grades were calculated using the upper bound of the 95% confidence interval on the percentage of correct answers. The percentages were then assigned to a letter-grade where A ($> 90.0\%$), B (80.0–89.9%), C (70.0–79.9%), D (60–69.9%), F (33.3–59.9%), and W (worse than random, $< 33.3\%$). These results show that Claude3.7, o3, and o4-mini

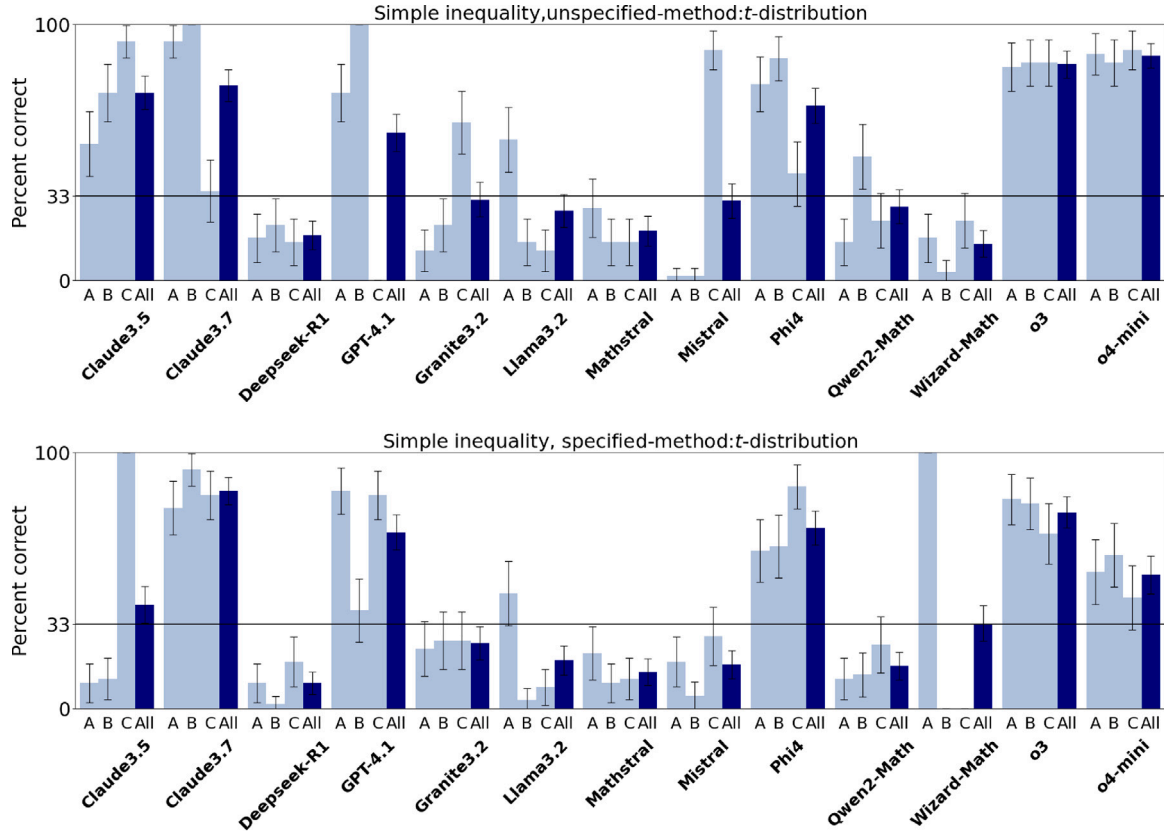


Fig. 3. Simple inequality unspecified-method and specified-method tests graded against t -distribution confidence intervals. The black line at 33% corresponds to random guessing.

Table 3

Grades for Simple Inequality tests. In the column headers, t refers to t -distribution solutions and B refers to bootstrapped solutions. DeepSeek[†] represents DeepSeek-R1-Distill-Qwen-32B.

Model	t	B	With method, t	With method, B
Claude3.5 *	C	C	F	C
Claude3.7 *	B	B	A	B
DeepSeek [†]	W	W	W	F
GPT-4.1 *	D	D	C	C
Granite3.2 - 8B	F	F	W	W
Llama3.2 - 1B	F	W	W	W
Mathstral - 7B	W	W	W	W
Mistral - 7B	F	W	W	W
o3 *	B	A	B	B
o4-mini *	A	A	F	B
Phi4 - 14B	C	C	C	F
Qwen2-Math - 7B	F	F	W	W
Wizard-Math - 7B	W	F	F	F

performed the best when computing basic statistical calculations of a simple dataset in the Simple Inequality tests.

In the Complex Inequality tests, we introduced higher degrees of arithmetic intensity and example-specific ontology, resulting in a dramatic loss of performance across the board (Table 4). These findings highlight that most models either failed or performed worse than random chance when tasked with greater arithmetic intensity UQ problems.

4.1. Comparisons of unspecified- and specified-method tests across UQ methods

The unspecified-method: t -distribution Simple Inequality test results revealed that o4-mini (92.6%), o3 (89.7%), and Claude3.7 (82.3%) performed the best. Wizard-Math (19.6%) scored the lowest overall.

When graded against bootstrapped confidence intervals, the top 5 performing models preserved their rankings except for o3, improving from a B grade to an A grade. In the specified-method: t -distribution Simple Inequality benchmark, Claude3.7 (90.2%), o3 (82.8%), GPT4.1 (75.7%), and o4-mini (59.5%) were the top performers. When graded against bootstrapped solutions o3 (89.3%) and Claude3.7 (78.8%) were the top performers, followed by o4-mini (82.3%), then GPT4.1 (78.8%) and Claude3.5 (78.8%), which both got the same score (Fig. A.5).

The unspecified-method Complex Inequality benchmarks for both UQ methods revealed that o3 (57.9%, 47.7%) and o4-mini (46.6%, 43.7%) performed the best. Mistral proved to be the most skeptical LLM by answering all prompts with “uncertain”. With both UQ methods, 10 models did better than random choice equal chance probability. Analysis of the specified-method Complex Inequality test revealed that o4-mini (86.9%), o3 (80.8%), Claude3.7 (67.2%), GPT4.1 (59.5%) and Claude3.5 (59.5%) performed the best when compared against t -distribution confidence intervals. All other models did significantly worse. With bootstrapped confidence intervals, both o3 (70.9%) and o4-mini (70.4%) dropped a letter grade (Fig. A.6).

Comparing the specified- and unspecified-methods in the Simple Inequality test, some models saw a decrease in performance (Claude3.5), but on average performance did not drastically change (Fig. 3). In Complex Inequality tests, most models saw a decrease in performance with the specified-methods tests, but the opposite is true for o4-mini, o3, GPT4.1, Claude3.5 and Claude3.7 (Fig. 4). Mistral, Wizard-Math, and Qwen2-Math all performed substantially better without the UQ method specified in the prompt. This result shows different model’s abilities to navigate decision-making based on general versus specific instructions in UQ calculations.

All models either failed or performed worse than random chance in the unspecified-method-specified-ontology tests. o3 (57.3%), o4-mini (56.7%), and Claude3.7 (52.3%) performed the best, but still received a

Table 4

Grades for Complex Inequality test. In the column headers, t refers to t -distribution solutions and B refers to bootstrapped solutions. DeepSeek[†] represents DeepSeek-R1-Distill-Qwen-32B.

Model	t	B	With method, t	With method, B	With ontology, t	With ontology, B
Claude3.5 *	F	F	F	F	F	F
Claude3.7 *	F	F	D	F	F	F
DeepSeek [†]	W	W	W	W	F	W
GPT-4.1 *	F	F	F	F	F	F
Granite3.2 - 8B	F	F	W	W	F	F
Llama3.2 - 1B	W	W	W	W	W	W
Mathstral - 7B	W	W	W	W	W	W
Mistral - 7B	F	F	W	W	F	F
o3 *	F	F	B	C	F	F
o4-mini *	F	F	B	C	F	F
Phi4 - 14B	F	F	F	W	W	W
Qwen2-Math - 7B	F	F	W	W	W	F
Wizard-Math - 7B	F	W	W	W	W	W

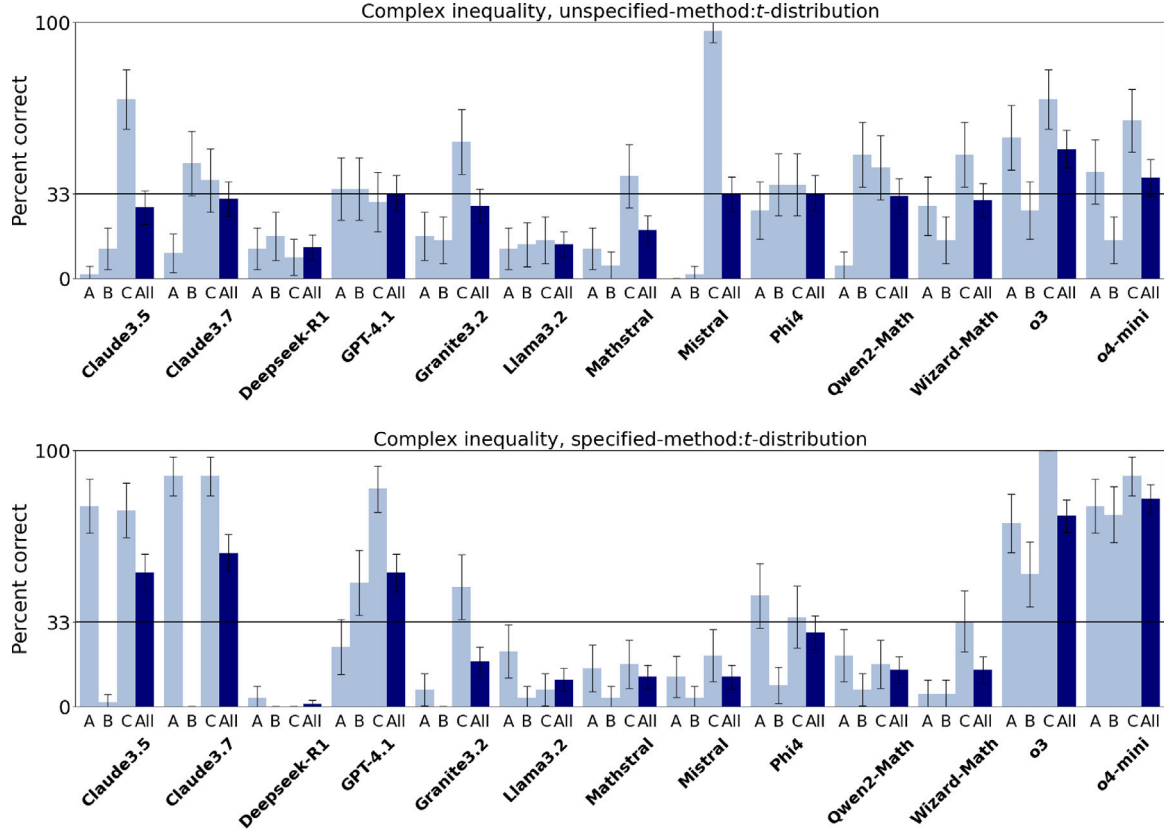


Fig. 4. Complex inequality unspecified-method and specified-method tests graded against t -distribution confidence intervals. The black line at 33% corresponds to random guessing.

failing grade (Fig. A.7). Models like Mistral and Granite3.2 were able to correctly answer “uncertain” with higher accuracy than other models, which implies that other models were less confident admitting uncertainty. Reducing the amount of information provided in the prompt and introducing example specific ontology affected the LLM’s ability to follow instructions and perform the necessary calculations to find the answer.

4.2. Limitations and further analyses

There are three discrete steps in the benchmarking algorithm: generate, run, and grade. During the ‘generate’ phase, questions are generated and saved as a blank exam. During the ‘run’ phase, the exams are given to an LLM and the responses are saved. The saved responses can be analyzed to reveal links between UQ methods stated by LLMs and their subsequent performance (details in Appendix A.1.2). While exhaustive analyses of this kind are beyond the scope of the manuscript,

we identified that some models mentioned a variety UQ methods (e.g., Welch’s t -test, χ -square test, Fisher’s exact test, the Cochran–Mantel–Haenszel test, etc.) and the link between the UQ methods purportedly used by the LLMs and LLM performance is not analyzed in this study.

While our benchmark demonstrations are limited to non-agentic LLMs, LLM-based agents could be readily assessed by the benchmarks with only minor to no modification required, depending on the form of the final output response of the LLM-based agents. We recognize that the benchmarks presented here focused on aleatoric (inherently random) sources of uncertainty and did not account for systematic uncertainty. UQ benchmarks that address combined random and systematic uncertainties are necessary because both sources of error are germane to ‘real world’ scientific observations.

Only one of our benchmarks, the bootstrapped Complex Inequality benchmark, measures uncertainty propagation (UP) performance. Other benchmarks that include measures of LLM performance at various UP methods such as Monte Carlo UP, the delta method, and

Bayesian credible intervals for UP are necessary. In all cases, we applied the most rudimentary UQ methods to the benchmark results because rudimentary UQ benchmarking is the first step towards more complex UQ benchmarks. Such methods should be eventually included in a more comprehensive UQ benchmark suite.

5. Conclusion

To our knowledge, the proposed benchmark suite is the first designed to assess a model's ability to perform the most fundamental UQ task: the evaluation of inequalities under uncertainty. It introduces a new scientifically rigorous objective for state-of-the-art models, continuing a pattern where such objectives have previously driven progress in AI performance (Pineau et al., 2021). The proposed benchmark suite provides a targeted assessment of fundamental UQ capabilities that any model intended for deployment on more complex, real-world UQ tasks should be able to master. Since the benchmarks are user-generated with randomization, they are unlikely to appear in LLM training data, can be easily scaled to reduce aleatoric uncertainty, are easily downloaded, and are extensible to include new prompt strategies (e.g., the addition of ontology to Simple Inequality benchmarks) and UQ methodologies (e.g., changing the percent confidence interval to 99%, the addition of myriad UP methods to Complex Inequality, etc.). The benchmark suite is also readily applicable to agents capable of writing and executing code for UQ tasks.

Our demonstration of the benchmark suite (Section 4) on 13 models indicates that, while a subset of models can exceed random-guessing performance in simple empirical inequality estimation, most models exhibit substantial performance degradation when faced with increased arithmetic complexity and/or the need to propagate uncertainty through multivariate probability functions. The performance of o3 on the Simple Inequality unspecified-method (Table 3) highlights the value of the benchmark suite: it supports the conclusion that, in the absence of explicit instruction, o3 tends to favor the use of bootstrapped confidence intervals over t -distribution confidence intervals when estimating an inequality under uncertainty without detailed *a priori* knowledge of data set idiosyncrasies. As we discussed in Section 3.1, we take a scientist-centric view of prompting: A trustworthy UQ assistant should be able to reliably perform basic UQ tasks in a manner that is robust with respect to prompt strategy. This standard, is supported by the performance of state-of-the-art models on Simple Inequality, where the latest Anthropic and OpenAI models performed reliably well (Table 3). We also found that, as the arithmetic complexity of the UQ task was increased from Simple Inequality to Complex Inequality, model performance decreased. While this result echoes prior observations that LLM accuracy on simple arithmetic degrades rapidly with increased digit length and/or compositional complexity (Frieder et al., 2023), we note that the benchmark suite is readily applicable to agents that can potentially overcome this shortcoming by writing and executing code. The demonstration of the proposed benchmark suite reveals that trustworthy UQ is not only an essential objective for scientific applications but also a challenging one, with significant progress still needed.

CRedit authorship contribution statement

Dorianis M. Perez: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Bryan E. Kaiser:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Ismael Boureima:** Resources, Data curation, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendices and supplementary material

A.1. Additional benchmark results

A.1.1. Model performances compared to random guessing

Figs. A.5, A.6, and A.7 are presented here for reference in the discussion of results in Section 4.

A.1.2. Analyses of model methodology

We present a wordcloud and related word frequency histogram for the Complex Inequality unspecified-method test on Wizard-Math (Fig. A.8). Each word was counted only once per response. Wizard-Math tended to provide more verbose responses with mention of specific statistical verbiage. This is a demonstration of future analysis of UQ methods employed by different LLMs to solve the questions in our benchmark suite.

A.2. Simple inequality benchmark details

A.2.1. Simple inequality uncertainty quantification

Student's t -distribution method: The sample standard deviation (SD) of the mean difference for the simple inequality benchmarks

$$\Delta_{xy} = \bar{X} - \bar{Y}, \quad (\text{A.1})$$

is

$$SD = \sqrt{\frac{\sum_{i=1}^{N_{xy}} (\Delta_{xy} - \bar{\Delta}_{xy})^2}{1 - N_{xy}}}, \quad (\text{A.2})$$

where overbars represent mean values

$$\bar{Z} = \frac{1}{N_z} \sum_{i=1}^{N_z} Z_i, \quad (\text{A.3})$$

and therefore $\bar{X}$ and $\bar{Y}$ (both of length $N_{xy} = 10$) are the means of vectors X and Y .

The difference is used for confidence interval (CI) estimation because two statistics may have overlapping 95% CIs but still have a statistically significant difference. This is because the distribution of the difference has its own standard error, which is usually smaller than the sum of individual errors (especially when samples are independent) (Cumming, 2009). The standard deviation is used to calculate the t -distribution confidence intervals

$$CI = \pm t_{\alpha/2} \frac{SD}{N_{xy}} \quad (\text{A.4})$$

where $t_{\alpha/2} \approx 1.96$ for the 95% confidence level (i.e., 95% of the area under a standard normal probability density lies between the 2.5th percentile and 97.5th percentile points) if the underlying distribution is a standard normal distribution and $N \geq 30$ (Wheeler, Ganji, Krishnan, & Thurow, 1996). Thus, the t -distribution estimate of the error of the mean (i.e., the most probable for a normal density) value of X is $\bar{X} \pm CI$.

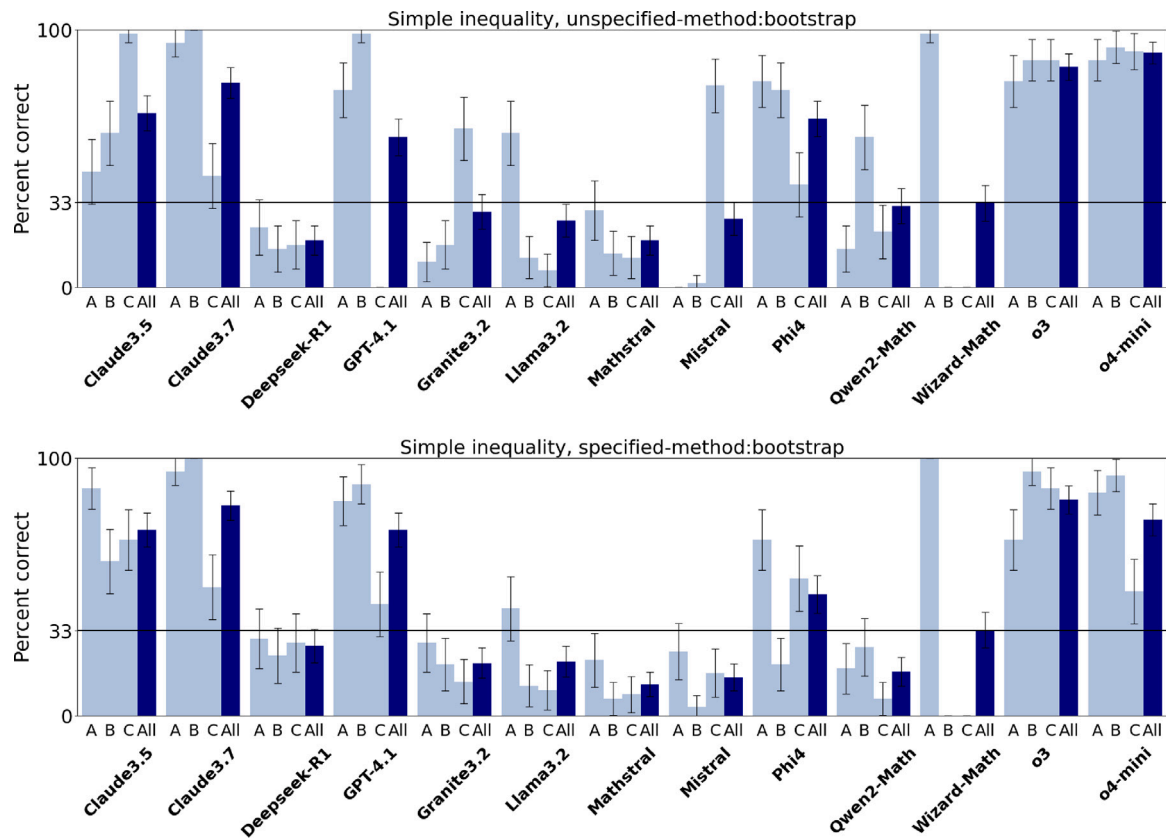


Fig. A.5. Simple inequality unspecified-method and specified-method tests graded against bootstrapped confidence intervals.

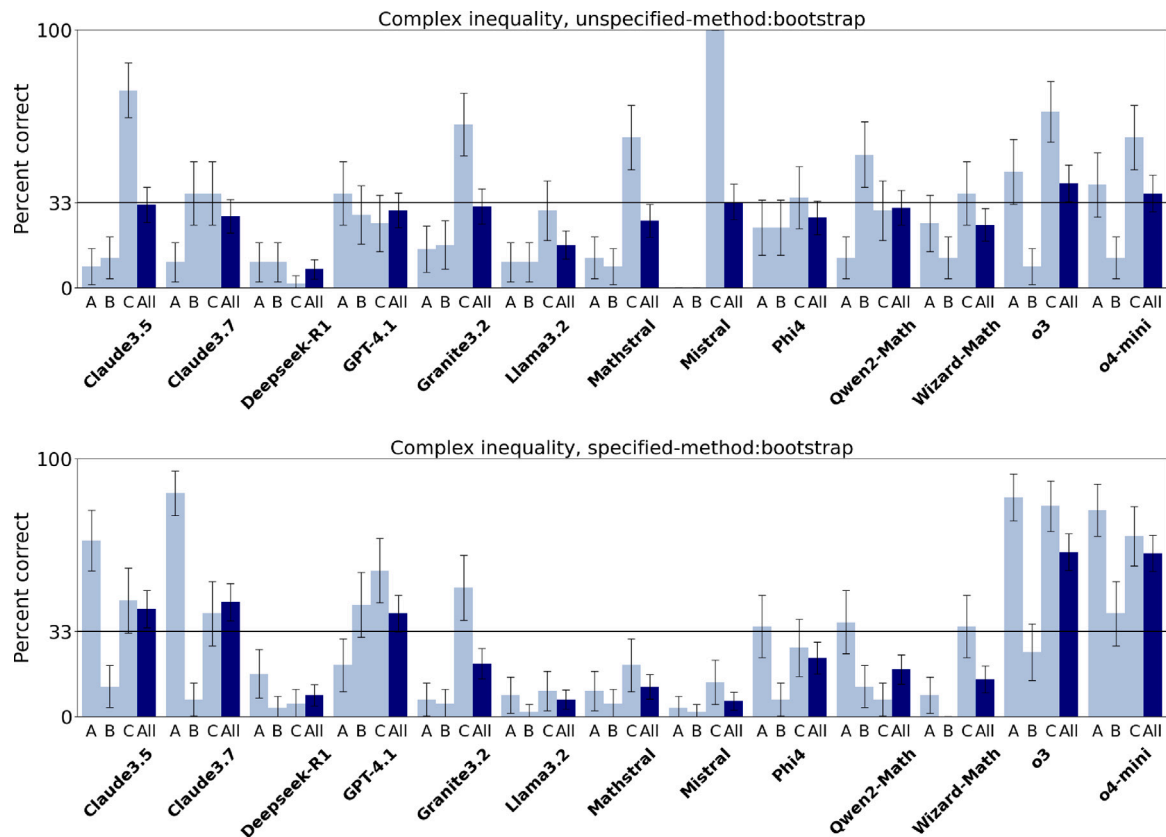


Fig. A.6. Complex inequality unspecified-method and specified-method tests graded against bootstrapped confidence intervals.

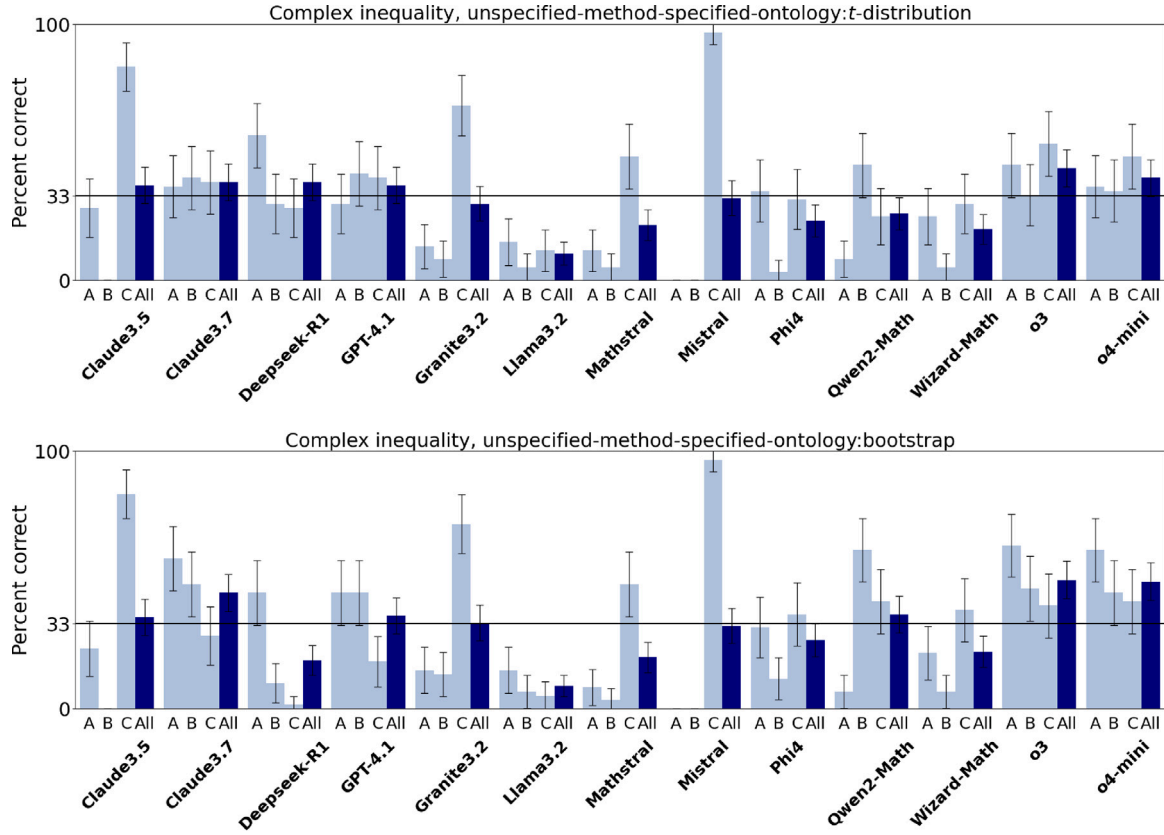


Fig. A.7. Complex inequality unspecified-method-specified-ontology tests graded against t -distribution and bootstrapped confidence intervals.

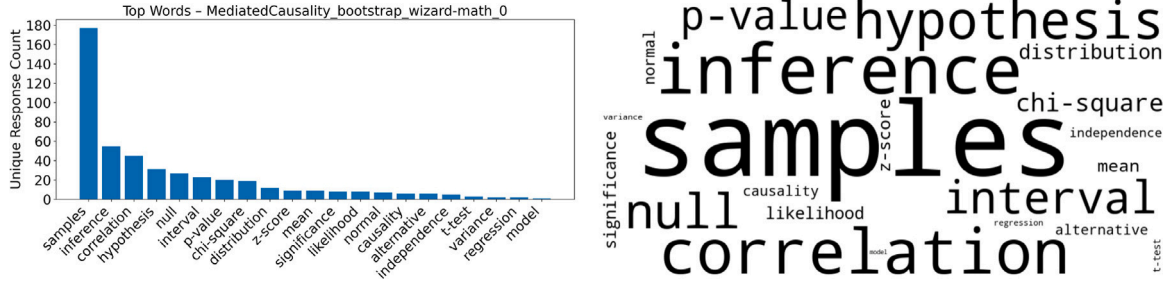


Fig. A.8. Word frequency histogram and wordcloud from the Complex Inequality unspecified-method test on Wizard-Math.

Bootstrap method: Bootstrapping is a simple non-parametric technique for estimating confidence intervals by resampling the observed data (Efron & Tibshirani, 1994). It is useful when the theoretical distribution of a statistic is complex or unknown. First, the data is resampled N_r times with replacement. Then, the statistic of interest (i.e., mean) is calculated of each of the N_r resampled datasets. Finally, the 2.5th and 97.5th percentiles of the N_r statistic of interest (i.e., mean) define the upper and lower bounds of the confidence interval. For every bootstrap benchmark presented here $N_r = 10^3$.

A.2.2. Simple inequality problem example

The Simple Inequality benchmark serves to test model performance when asked to do a base level statistical calculation of inequalities under uncertainty, given two vectors of normally distributed data. Each test was created with constraints specified in Section 3. Given two vectors of equal length ($\text{len}(X) = \text{len}(Y) = N_{xy} = 10$)

$$X = [-2.955, -1.173, -2.952, -0.423, -0.392, -1.618, -0.629, -3.724, -1.632, 0.623], \quad (A.5)$$

$$Y = [0.179, 2.428, -2.286, 0.728, -2.949, -3.940, 3.129, 2.984, 1.789, 2.687]. \quad (A.6)$$

their means are computed using Eq. (A.3) such that $\bar{X} = -1.488$ and $\bar{Y} = 0.475$. The absolute mean difference $|\Delta| = |\bar{X} - \bar{Y}| = 1.962$ informs vector pair generation, such that an equal number of vector pairs are generated for mean differences in bins. In this case, this example is a medium-difficulty question. The sample standard deviation and 95% confidence intervals are computed around the mean difference, Δ , according to Eqs. (A.2) and (A.4). The vectors were generated such that the sample standard deviations have random magnitudes so long as resulting solution is the desired one. In this example, 95% confidence intervals for Δ are

$$\Delta_{xy} = -1.962 \pm 1.843 \quad (A.7)$$

Thus, the solution for this example is $B: X < Y$ within 95% confidence because $-1.962 + 1.843 = -0.119 < 0$.

To determine whether the first list of numbers is larger than the second list, the lower boundary of the confidence interval has to be larger than 0 (answer A). If the upper boundary of the confidence

Table A.5

The maximum PSRDs by question category for 10 separate generations of the Simple Inequality t -distribution benchmark. The maximum PSRD of $\hat{\Delta}$ and $\hat{\sigma}_{\Delta}$ is obtained by finding the maximum difference between any two benchmark's average Δ and average σ_{Δ} , respectively. These data show the robust quantitative similarity of multiple generations of the same benchmark.

Solution	Difficulty	max(PSRD($\hat{\Delta}, \hat{\Delta}^T$))	max(PSRD($\hat{\sigma}_{\Delta}, \hat{\sigma}_{\Delta}^T$))
A	Easy	0.11%	0.14%
A	Medium	0.02%	0.09%
A	Hard	0.02%	0.10%
B	Easy	0.07%	0.10%
B	Medium	0.04%	0.06%
B	Hard	0.03%	0.06%
C	Easy	1.86%	0.06%
C	Medium	1.80%	0.06%
C	Hard	1.82%	0.03%

interval is less than zero, than list 2's mean is larger than list 1's mean (answer B). If neither of these is true, than the answer is uncertain (answer C). In this example, since the confidence interval is less than 0, we'd have an answer of B meaning that the second list has a statistically significantly higher mean than the first list.

A.2.3. Simple inequality reproducibility verification

Benchmark lengths of $N_q = 180$ questions was chosen because the magnitudes of the most probable difference between the two variables are binned into three magnitude bins and because the multiple choice format has three choices (e.g., $X > Y$, $X < Y$, uncertain, all within 95% confidence). Therefore, each set of correct answer choices must be evenly split ($N_q/3$, $N_q/3$, $N_q/3$) and within each of the answer choices the split of magnitude bins is even ($N_q/9$, $N_q/9$, $N_q/9$).

We use the percent symmetric relative difference (PSRD)

$$\text{PSRD}(a, b) = 100 \frac{|a - b|}{(a + b)/2} \quad (\text{A.8})$$

to compare the statistics of the separate generations of the benchmarks because the SRD is scale-invariant (normalized by the magnitude of the values), symmetric ($\text{PSRD}(a, b) = \text{PSRD}(b, a)$), bounded ($\text{PSRD}(a, b) \geq 0$), and unitless.

$\hat{\Delta}$ is the sample mean of the magnitude difference Δ for a single benchmark, where the averaging is over all questions with the same correct answer A, B, or C for the same difficulty level. $\hat{\sigma}_{\Delta}$ is the sample standard deviation of the magnitude difference Δ for a single benchmark, taken over all questions with for the same correct answer A, B, or C for the same difficulty level.

We generated the Simple Inequality t -distribution 10 times to obtain a set of $\hat{\Delta}$ and $\hat{\sigma}_{\Delta}$ values, such that:

$$\vec{\Delta} = [\hat{\Delta}_1, \hat{\Delta}_2, \dots, \hat{\Delta}_{10}], \quad (\text{A.9})$$

$$\vec{\sigma}_{\Delta} = [\hat{\sigma}_{\Delta,1}, \hat{\sigma}_{\Delta,2}, \dots, \hat{\sigma}_{\Delta,10}], \quad (\text{A.10})$$

and compared the differences. These are shown in Table A.5.

A.3. Complex inequality benchmark details

A.3.1. Calculation of 'do' probabilities from frequency tables

The 'do' probabilities $P(Y = 1 | \text{do}(X = 1))$ and $P(Y = 1 | \text{do}(X = 0))$ can be calculated using Eq. (1), a frequency table (Table 2), and the following relations for the relevant marginal, joint, and conditional probabilities (Pearl, 2009):

$$N_s = \sum_{i=1}^8 n_i, \quad (\text{A.11})$$

$$P(X = 0) = \frac{n_1 + n_3 + n_2 + n_4}{N_s}, \quad (\text{A.12})$$

$$P(X = 1) = \frac{n_5 + n_7 + n_6 + n_8}{N_s}, \quad (\text{A.13})$$

$$P(Z = 0 | X = 1) = \frac{n_7 + n_5}{n_8 + n_6 + n_7 + n_5}, \quad (\text{A.14})$$

$$P(Z = 1 | X = 1) = \frac{n_8 + n_6}{n_8 + n_6 + n_7 + n_5}, \quad (\text{A.15})$$

$$P(Z = 0 | X = 0) = \frac{n_3 + n_1}{n_4 + n_2 + n_3 + n_1}, \quad (\text{A.16})$$

$$P(Z = 1 | X = 0) = \frac{n_4 + n_2}{n_4 + n_2 + n_3 + n_1}, \quad (\text{A.17})$$

$$P(Y = 1 | X = 0, Z = 0) = \frac{n_3}{n_1 + n_3}, \quad (\text{A.18})$$

$$P(Y = 1 | X = 1, Z = 0) = \frac{n_7}{n_5 + n_7}. \quad (\text{A.19})$$

A.3.2. Complex inequality uncertainty quantification

The final 'do' probabilities are calculated from marginal, joint, and conditional probabilities that are all calculated from the same frequency table. Since each marginal, joint, and conditional probability may have its own variance, the most conservative UQ methods for the Complex Inequality benchmarks involve accounting for Uncertainty Propagation (UP) from the marginal, joint, and conditional probabilities to the final 'do' probabilities.

Student's t -distribution method (no UP): The sample standard error for the difference of the probabilities

$$\Delta_P = P(Y = 1 | \text{do}(X = 1)) - P(Y = 1 | \text{do}(X = 0)) \quad (\text{A.20})$$

is

$$\text{SE} = \sqrt{\frac{P_{d0}(1 - P_{d0}) + P_{d1}(1 - P_{d1})}{N}}, \quad (\text{A.21})$$

which is used to calculate the t -distribution confidence intervals

$$\text{CI} = \pm t_{\alpha/2} \cdot \text{SE}_{xy}, \quad (\text{A.22})$$

where $t_{\alpha/2} \approx 1.96$ for the 95% confidence level, meaning that 95% of the area under a normal curve lies between the 2.5th percentile and 97.5th percentile points and thus, the probability is 95% given the data at hand Brayer (2018).

Bootstrap method (with UP): In the bootstrap method the frequency table data is expanded into individual observations which can be resampled. The expanded data is resampled with replacement N_r times and the probability difference is computed each time. This yields an empirical distribution of the probability difference $P_{d0} - P_{d1}$ which is then used to calculate the mean and the 2.5th and 97.5th percentiles to form the 95% confidence interval. This is statistically equivalent to UP through the products of probabilities in Eq. (1) in the limit as $N_s \rightarrow \infty$ because those probabilities are all derived from the same underlying frequency structure.

A.3.3. Complex inequality problem example

The objective of this benchmark is to assess performance on assessing empirical inequalities under uncertainty when the inequality requires several steps of calculations to obtain. Table A.6 is equivalent to the problem described in 3.3.3 Example: Smoking and the Genotype Theory of Pearl (Pearl, 2009). LLM prompts contain the information in Table A.6, as shown in 1. Given the frequency table (Table A.6) and the marginal, joint, and conditional probabilities in Appendix A.3.1 the relevant joint probabilities are

$$P(X = 0, Z = 0) = 0.475 \text{ (Probability a sample does not smoke, no tar)}, \quad (\text{A.23})$$

$$P(X = 1, Z = 0) = 0.025 \text{ (Probability a sample does smoke, no tar)}, \quad (\text{A.24})$$

$$P(X = 0, Z = 1) = 0.025 \text{ (Probability a sample does not smoke, tar)}, \quad (\text{A.25})$$

$$P(X = 1, Z = 1) = 0.475 \text{ (Probability a sample does smoke, tar)}, \quad (\text{A.26})$$

Table A.6

Frequency table of observations of X , Y , and Z that is equivalent to the problem described in 3.3.3 *Example: Smoking and the Genotype Theory of Pearl* (Pearl, 2009). $X = 1$ represents smoking, $Z = 1$ represents the presence of tar in lungs, and $Y = 1$ represents lung cancer. The total number of samples is $N_s = 10^5$. In this synthetic example, smoking does not cause lung cancer.

X	Y	Z	Samples
0	0	0	42750
0	0	1	2375
0	1	0	4750
0	1	1	125
1	0	0	250
1	0	1	7125
1	1	0	2250
1	1	1	40375

and the relevant marginals are

$$P(X = 1) = P(X = 1, Z = 0) + P(X = 1, Z = 1) = 0.50, \quad (\text{A.27})$$

$$P(X = 0) = 1 - P(X = 1) = 0.50. \quad (\text{A.28})$$

and the relevant conditional joint probabilities are

$$\begin{aligned} P(Y = 1|X = 0, Z = 0) \\ = 0.10 \text{ (Probability of cancer given no smoking, no tar),} \end{aligned} \quad (\text{A.29})$$

$$\begin{aligned} P(Y = 1|X = 1, Z = 0) \\ = 0.90 \text{ (Probability of cancer given smoking, no tar),} \end{aligned} \quad (\text{A.30})$$

$$\begin{aligned} P(Y = 1|X = 0, Z = 1) \\ = 0.05 \text{ (Probability of cancer given no smoking, tar),} \end{aligned} \quad (\text{A.31})$$

$$\begin{aligned} P(Y = 1|X = 1, Z = 1) \\ = 0.85 \text{ (Probability of cancer given smoking, tar),} \end{aligned} \quad (\text{A.32})$$

and the relevant conditional probabilities are

$$P(Z = 0|X = 1) = \frac{P(X = 1, Z = 0)}{P(X = 1)} = 0.05, \quad (\text{A.33})$$

$$P(Z = 1|X = 1) = \frac{P(X = 1, Z = 1)}{P(X = 1)} = 0.95, \quad (\text{A.34})$$

$$P(Z = 0|X = 0) = \frac{P(X = 0, Z = 0)}{P(X = 0)} = 0.95, \quad (\text{A.35})$$

$$P(Z = 1|X = 0) = \frac{P(X = 0, Z = 1)}{P(X = 0)} = 0.05. \quad (\text{A.36})$$

Substitution of the marginal, joint, and conditional probabilities from Eqs. (A.23)–(A.36) into Eq. (1) yields:

$$\begin{aligned} P(Y = 1|\text{do}(X = 1)) = \\ P(Z = 0|X = 1)[P(Y = 1|X = 0, Z = 0)P(X = 0) \\ + P(Y = 1|X = 1, Z = 0)P(X = 1)] \\ + P(Z = 1|X = 1)[P(Y = 1|X = 0, Z = 1)P(X = 0) \\ + P(Y = 1|X = 1, Z = 1)P(X = 1)] \\ = 0.05[0.01 \cdot 0.5 + 0.90 \cdot 0.5] + 0.95[0.050 \cdot 0.5 + 0.85 \cdot 0.50] \\ = 0.4525, \end{aligned} \quad (\text{A.37})$$

and

$$\begin{aligned} P(Y = 1|\text{do}(X = 0)) = \\ P(Z = 0|X = 0)[P(Y = 1|X = 0, Z = 0)P(X = 0) \\ + P(Y = 1|X = 1, Z = 0)P(X = 1)] \\ + P(Z = 1|X = 0)[P(Y = 1|X = 0, Z = 1)P(X = 0) \\ + P(Y = 1|X = 1, Z = 1)P(X = 1)] \\ = 0.95[0.01 \cdot 0.5 + 0.90 \cdot 0.5] + 0.05[0.050 \cdot 0.5 + 0.85 \cdot 0.50] \\ = 0.4975. \end{aligned} \quad (\text{A.38})$$

Table A.7

The maximum percent Wasserstein distances between 10 separate generations of the Complex Inequality t -distribution benchmark. The distances are computed for both the means of the normalized frequency vectors μ and for the standard deviations σ of the normalized frequency vectors. These data show the robust quantitative similarity of multiple generations of the same benchmark.

Solution	Difficulty	max[% $W(\mu, \mu)$]	max[% $W(\sigma, \sigma)$]
A	Easy	15.6%	20.8%
A	Medium	15.6%	19.1%
A	Hard	19.8%	25.7%
B	Easy	20.6%	27.8%
B	Medium	19.6%	35.5%
B	Hard	14.7%	21.2%
C	Easy	33.2%	35.9%
C	Medium	26.2%	28.9%
C	Hard	17.3%	27.8%

which are identical to the result in Pearl (Pearl, 2009). Thus the difference is $\Delta = P(Y = 1|\text{do}(X = 1)) - P(Y = 1|\text{do}(X = 0)) = -0.045$ which indicates that the data in Frequency Table 2 refutes the hypothesis that smoking causes cancer.

To calculate the 95% confidence intervals on the difference Δ (no uncertainty propagation) we use Eq. (A.22) with $N = 10^5$ from the sum of the rightmost column in Frequency Table 2. In this example, the answer for the difference between the ‘do’ probabilities is

$$\Delta_P = -0.045 \pm 0.002. \quad (\text{A.39})$$

Since $-0.045 + 0.002 = -0.043 < 0$, the evidence in Frequency Table 2 supports the claim that smoking does not cause cancer within 95% confidence. Eq. (A.39) corresponds to answer B: $P(Y = 1|\text{do}(X = 1)) < P(Y = 1|\text{do}(X = 0))$ within 95% confidence.

A.3.4. Complex inequality reproducibility verification

First we get the frequency vectors (the number of samples in the rightmost column of Table 2) for a single difficulty and solution. Each vector is normalized by the total number of samples, N_s (Table 2). Then, the mean vector μ and the standard deviation σ vector (each of length 8 because the combination of three binary variables is $2^3 = 8$) are obtained for each difficulty and solution. These vectors are combined into arrays of 10 vectors, one for each generated benchmark:

$$\mu = \{\mu_1, \dots, \mu_{10}\}, \quad (\text{A.40})$$

$$\sigma = \{\sigma_1, \dots, \sigma_{10}\}, \quad (\text{A.41})$$

where both have 10 rows and 8 columns. Next, the percent Wasserstein (% W) distances between the mean and standard deviation frequency vectors (% $W(\mu, \mu)$ and % $W(\sigma, \sigma)$) are computed to compare different generated benchmarks,

$$W(x, y) = \frac{1}{8} \sum_{i=1}^8 |x_i - y_i| \quad (\text{A.42})$$

$$\%W(\mathbf{x}, \mathbf{y}) = 100 \frac{W(\mathbf{x}, \mathbf{y})}{\max[x_i, y_j] - \min[x_i, y_j]}, \quad i = \{1, \dots, 8\}, j = \{1, \dots, 8\}, \quad (\text{A.43})$$

such that the resulting set of % W is an array with 10 rows and 8 columns. Finally, we take the maximum values across the rows (generated benchmarks) to show the resulting maximum percent Wasserstein distances in Table A.7 for each difficulty and solution when calculated over 10 different generations of the Complex Inequality t -distribution benchmark. The % W is useful for quantifying the variability of generated frequency tables because it scales the distance by the spread of the data, makes the percent distance invariant to overall scale, and is useful for reporting differences in relative terms, especially for non-centered, non-Gaussian, or arbitrary-range vectors. Table A.7 indicates that, over

10 generations of the Complex Inequality t -distribution benchmark, the mean of the normalized frequency vector varies by a maximum of 33% and the standard deviation of the normalized frequency vector varies by a maximum of 36%. This conservative estimation of generation variability indicates that, in addition to the fixed set of problems generated for each solution and difficulty type, the generated frequency tables for each question type are statistically similar from one generation to the next.

Data availability

The link to data/code is available in the manuscript.

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